WORKED SOLUTION

STEP III 1992, Question 12

Mechanics • Conservation of Momentum

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Read the question before you go on. It is in the STEP Support Programme's question database:

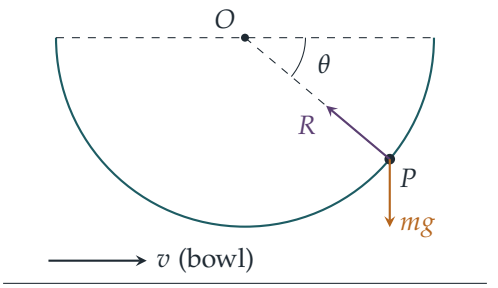
step.maths.org/questions/92-s3-q12

The key idea

Nothing horizontal acts on the bowl-and-particle system from outside, and nothing does work except gravity, so we have two conservation laws: horizontal momentum and energy. Momentum links the bowl's speed to $\dot{\theta}$; energy then gives v^2. For the reaction, the cleanest route is to work in the bowl's (accelerating) frame, where the particle simply moves in a circle about O, and to remember that the reaction is also the only horizontal force on the bowl.

Solution

Model. The bowl is smooth, so the reaction R on P acts along the normal, that is along PO. The carriage is light and the table smooth, so the only horizontal force on the bowl (with its carriage) is the horizontal component of the particle's push. Take P to start at the right-hand end of the rim, so θ increases from 0 as P slides down; the bowl then moves to the right.



Conservation of horizontal momentum

Let the bowl's horizontal position be X, so $\dot{X} = v$. Relative to O, the particle is at $(a\cos\theta,\ -a\sin\theta)$, so its position is $(X + a\cos\theta,\ -a\sin\theta)$ and its velocity is

$$(v - a\dot{\theta}\sin\theta,\ -a\dot{\theta}\cos\theta).$$

There is no external horizontal force on the system, and it starts at rest, so the total horizontal momentum stays zero:

$$2mv + m(v - a\dot{\theta}\sin\theta) = 0 \quad \Longrightarrow \quad 3v = a\dot{\theta}\sin\theta,$$

as required. Notice that the particle's horizontal velocity is then $v - 3v = -2v$: it moves left at twice the bowl's speed, as it must to cancel the bowl's momentum $2mv$.

Energy

The reaction does no net work (it is internal and the surface is smooth), and the table does no work, so energy is conserved. When OP is at angle θ the particle has fallen $a\sin\theta$. Using $a\dot{\theta} = 3v/\sin\theta$, the particle's vertical speed is $a\dot{\theta}\cos\theta = 3v\cot\theta$, so

$$\begin{aligned} mga\sin\theta &= \tfrac{1}{2}(2m)v^2 + \tfrac{1}{2}m[(2v)^2 + 9v^2\cot^2\theta] \\ &= mv^2\left(3 + \tfrac{9}{2}\cot^2\theta\right). \end{aligned}$$

Multiplying by $2\sin^2\theta/m$:

$$2ga\sin^3\theta = v^2(6\sin^2\theta + 9\cos^2\theta) = v^2(9 - 3\sin^2\theta),$$

so

$$v^2 = \frac{2ga\sin^3\theta}{3(3 - \sin^2\theta)},$$

which is the second result.

The reaction

Ask yourself The bowl is accelerating. How do we write down the particle's equation of motion along PO?

Use two equations. First, for the bowl: the particle pushes on it with force R directed from O to P, whose horizontal component is $R\cos\theta$ to the right. So the bowl's acceleration A satisfies

$$2mA = R\cos\theta. \tag{1}$$

Second, for the particle, work in the bowl's frame. There the particle moves on a circle of radius a about O with speed $a\dot{\theta}$, and as well as R and mg it feels a fictitious force mA to the left. Resolving towards O (direction $(-\cos\theta, \sin\theta)$), gravity contributes $-mg\sin\theta$ and the fictitious force contributes $+mA\cos\theta$, so

$$R - mg\sin\theta + mA\cos\theta = ma\dot{\theta}^2. \tag{2}$$

(Equivalently, in the ground frame the particle's acceleration is the bowl's acceleration A plus its circular acceleration relative to the bowl; resolving that along PO gives the same equation.)

From the first two results,

$$a\dot{\theta}^2 = \frac{9v^2}{a\sin^2\theta} = \frac{6g\sin\theta}{3-\sin^2\theta}.$$

Substituting $mA = \frac{1}{2}R\cos\theta$ from (1) and collecting the terms in R:

$$R\left(1+\tfrac{1}{2}\cos^2\theta\right) = mg\sin\theta + \frac{6mg\sin\theta}{3-\sin^2\theta} = mg\sin\theta\,\frac{9-\sin^2\theta}{3-\sin^2\theta}.$$

Since $1+\frac{1}{2}\cos^2\theta = \frac{1}{2}(3-\sin^2\theta)$,

$$\boxed{R = \frac{2mg\sin\theta\,(9-\sin^2\theta)}{(3-\sin^2\theta)^2} = \frac{2mg\sin\theta\,(8+\cos^2\theta)}{(2+\cos^2\theta)^2}.}$$

Checks

- At the start, $\theta = 0$ gives $R = 0$, as it should: the particle is at rest at the rim with nothing pressing it into the bowl.
- At the bottom, $\theta = \frac{1}{2}\pi$: the bowl's acceleration is zero (by (1)), $a\dot{\theta}^2 = 3g$, and $R - mg = 3mg$, so $R = 4mg$, matching the formula $2mg\cdot 8/2^2$.
- If the bowl were fixed (infinitely heavy), the usual result would be $v_P^2 = 2ga\sin\theta$ and $R = 3mg\sin\theta$; the recoiling bowl makes R smaller near the rim (about $2mg\sin\theta$ rather than $3mg\sin\theta$ for small θ, as the bowl moves away from P) but larger at the bottom ($4mg$ rather than $3mg$).
- Stepping the full equations of motion numerically (with $m = 1.3$, $a = 0.7$, $g = 9.81$) reproduced $3v = a\dot{\theta}\sin\theta$, the formula for v^2 and the formula for R to within 10^{-7} all the way from the rim to the bottom.

Common slips

- Writing $R - mg\sin\theta = ma\dot{\theta}^2$, as if the bowl were fixed. The bowl accelerates, so the particle's acceleration is not just its circular acceleration relative to the bowl.
- Using $\frac{1}{2}m(a\dot{\theta})^2$ for the particle's kinetic energy: $a\dot{\theta}$ is its speed relative to the bowl, not relative to the table.

Beyond STEP

This is a two-degree-of-freedom system, and the tidiest systematic treatment is Lagrangian mechanics: write the kinetic and potential energies in terms of X and θ, and

the momentum conservation drops out because X does not appear in the Lagrangian (it is a "cyclic" coordinate). The fictitious force we used is the simplest case of mechanics in a non-inertial frame, which first-year courses extend to rotating frames with centrifugal and Coriolis forces. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both non-inertial frames and systems of interacting bodies very well.

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